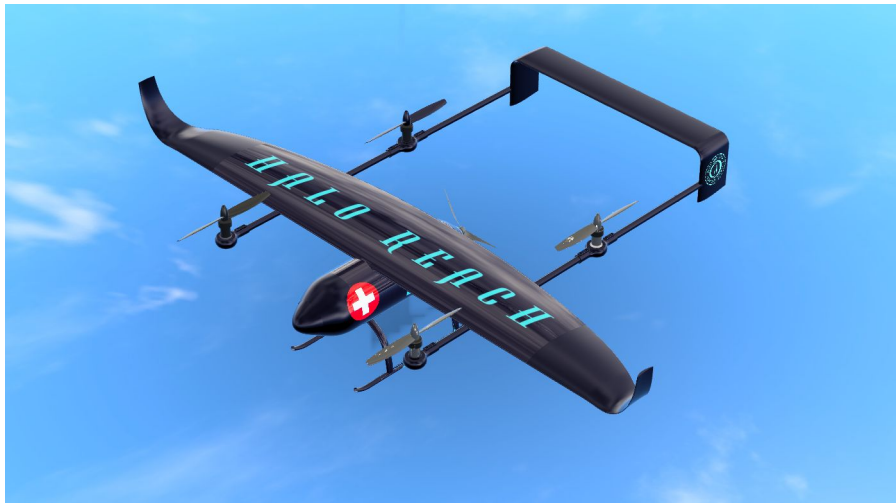

HALO REACH: Drag Analysis Methodology and Validity

UAS for Disaster Relief - AE3 2024



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Abstract

This report outlines the initial sizing and drag performance of the REACH UAV. A constraint diagram is used for the former, which found the landing zone area and cruise conditions to be the most constraining cases. The drag performance is analysed using both the equivalent skin friction and component build up methods, which are then compared and discussed. Computational Fluid Dynamics software is also used to predict drag, and this is first validated for a 2D aerofoil against experimental data. Turbulence modelling is also discussed and reviewed, and later the computational and analytical methods are reviewed, and possible improvements and refinements for later iterations of the aircraft discussed.

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Nomenclature

(L/D)	Lift-to-Drag Ratio
$(T/W)_0$	Takeoff Thrust to Weight Ratio
α	Angle of Attack
α_0	Zero-Lift Angle of Attack
η_h	Tailplane Efficiency Factor
Λ	Sweep
μ	Dynamic Viscosity
ρ	Density
σ	Propeller Solidity
A	Area
AR	Aspect Ratio
c	Chord Length
C_D	Drag Coefficient
C_L	Lift Coefficient
C_{D0}	Parasitic Drag Coefficient
C_{fe}	Equivalent Skin Friction Coefficient
C_f	Skin Friction Coefficient
$C_{L\alpha}$	Lift Curve Slope
D	Drag
d	Diameter
e	Oswald Efficiency
f	Fineness Ratio = l/d
FF	Form Factor
i_w	Wing Incidence
K	A Constant Value
L	Lift
l	Component Characteristic Length
M	Mach Number
Q_c	Interference Factor
Re	Reynolds Number
S_h	Tailplane Planform Area

S_{ref}	Wing Planform Area
S_{wet}	Wetted Surface Area
T	Thrust
t	Thickness
V	Velocity
W	Weight
W_0/S_{ref}	Wing Loading
x	Chordwise Position
CAD	Computer Aided Design
CFD	Computational Fluid Dynamics
HALO REACH	Highly Aerodynamic Light Operation Remote Emergency Aid & Crisis Help
NACA	National Advisory Committee for Aeronautics
RANS	Reynolds Averaged Navier Stokes
UAV	Uncrewed Aerial Vehicle
VTOL	Vertical Take-Off and Landing

1 Introduction

The aerodynamics team for the REACH UAV worked on initial sizing, wing and propeller design, and analytical and computational analysis of the aircraft as well as individual components. This report details the initial sizing process using a constraint diagram, then goes on to complete a theoretical analysis of the aircraft's parasitic and lift induced drag. This is later validated against CFD analysis in Star-CCM, which itself is validated using a 2D aerofoil against experimental data. The available turbulence models were also tested against wind tunnel data, and this report aims to discuss whether both CFD and analytical methods are a good fit for the drag modelling of a UAV with complex geometry, along with what could be improved about these methods for later iterations.

The driving force of this report is not wholly to gain an accurate value of the aircraft's drag at different mission segments, but also to review these analyses for the long-term project of HALO REACH. This will allow the project to continue improving performance throughout research and development stages. The full workload and task delegation of the aerodynamics team can be seen in Appendix B.

2 Constraints

For the initial sizing of the aircraft, it is necessary to first define the wing loading W_0/S_{ref} and the thrust-to-weight ratio $(T/W)_0$, from which the wing and propulsion requirements can be obtained [1]. This involves plotting these variables against each other at each mission segment with varying weight as fuel is used up. The aircraft weight estimates calculated by the Systems and Propulsion team were used for this purpose, and the constraint diagram in Figure 1 was drawn up following the method outlined by Raymer [1].

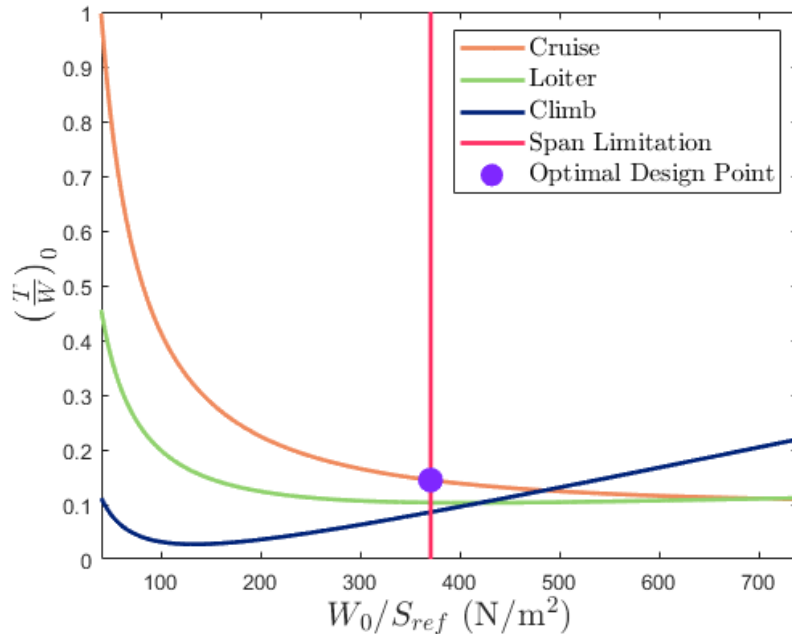


Figure 1. The constraint diagram for the mission, showing the optimal design point of $S_{ref} = 4.325 \text{ m}^2$, $(T/W)_0 = 0.144$. $(T/W)_0$ for VTOL was separately estimated at 1.04.

The most constraining aspects of the mission were the 1.25 hour cruise and the inherent wingspan limitation - the latter coming from the requirement to land in a 6 by 6 metre square. This naturally limited the wingspan, so by selecting the wing's aspect ratio, the wing planform area S_{ref} was also constrained. Hence the design point was selected to minimise $(T/W)_0$ and W_0/S_{ref} , which were then

used for engine and wing design respectively. The span constraint being a highly limiting factor meant that, later in the design process, winglets were added to the wing to increase effective span. Note that the VTOL mission stages were omitted in the diagram, as the wing is not used as a lifting surface in these cases, in addition to the pusher propeller being inactive. The design point for the VTOL rotors was done by simply balancing force and weight.

3 Analytical Drag Estimation

As the aircraft has two distinct orientations in VTOL and cruise/loiter, separate drag estimates must be made for each of these cases. As it is highly difficult to optimise the aircraft geometry for both cases, it was chosen to minimise drag during loiter (since it is by far the longest segment and therefore will have the larger impact on fuel consumption) and simply increase the thrust produced by the rotors to compensate for VTOL drag.

Multiple sources of drag are taken into account, though due to our low maximum Mach number of 0.13, drag divergence (and thus wave drag) present in the transonic regime does not occur as the aircraft will never exceed subsonic speeds (this was also ensured to be the case at the wingtips via calculation), leaving the two contributors to drag as parasite drag (from skin friction, form and interference drag) and lift induced drag (due to the lift vector not being perfectly perpendicular to the direction of travel).

3.1 Equivalent Skin Friction Method

The equivalent skin friction coefficient C_{fe} was used to estimate parasitic drag early on in the design cycle, as it requires only the wetted and planform areas of the wing to be known (this would later be refined as in the next section) using the method in Equation 1 [1]. C_{fe} was estimated as 0.0055 [2].

$$C_{D0} = C_{fe} \frac{S_{wet}}{S_{ref}} \quad (1)$$

The values used and the resulting parasite drag are listed in Table 2. This was expected to be an underestimate for C_{D0} , as empirical methods do not take into account the specifics of our aircraft such as the rotors which will be fully exposed during cruise.

Table 2. Optimised stringer and rib values

Variable	Value
C_{fe}	0.0055
S_{ref}	4.325 m ²
S_{wet}	14.402 m ²
C_{D0}	0.0183

3.2 Component Build Up Method

Once more geometry was finalised, the more accurate component build up method could be used [1]. This, naturally, builds up a drag profile from all of the components of the aircraft, giving a value for each component's contribution to the total drag. This allowed for the refinement of the geometry of components with excessive parasite drag. By calculating the values in Equation 8, the total C_{D0} can be found.

The skin friction coefficient C_{fc} is wholly dependent on the local Reynolds number, and is calculated as in Equation 2, where l is the characteristic length of the component (for example the mean aerodynamic chord for an aerofoil). C_{fc} was then calculated by use of Equation 3 for streamlined components in laminar flow, and Equation 4 for bluff components in turbulence.

$$Re = \frac{\rho V l}{\mu} \quad (2)$$

$$C_f = 1.328/\sqrt{Re} \quad (3) \quad C_f = \frac{0.455}{(\log_{10} Re)^{2.58} (1 + 0.144 M^2)^{0.65}} \quad (4)$$

Further corrections had to be made for the drag due to pressure - this is done by use of a form factor, FF , using empirical relations derived from experimental data. The form factor for the fuselage was estimated using Equation 6 and that for the wing and tail was found using Equation 5.

$$FF = \left[1 + \frac{0.6}{(x/c)_m} \left(\frac{t}{c} \right) + 100 \left(\frac{t}{c} \right)^4 \right] \left[1.34 M^{0.18} (\cos \Lambda_m)^{0.28} \right] \quad (5)$$

$$FF = 1 + \frac{60}{f^3} + \frac{f^2}{400} \quad (6)$$

Another contributor to parasitic drag is interference drag from components in close proximity, Q . Its value for each component was simply found empirically [1]. The drag due to the propeller during cruise and the rotors during VTOL were also considered, using Equation 7, in which the solidity σ was determined to be equal to the number of blades ($= 3$) divided by the blade aspect ratio ($= 8.8$) multiplied by pi.

$$C_{D\pi} = 0.1 \sigma A_{propellerdisk} \quad (7)$$

'Miscellaneous' drag sources from non-streamlined components are also essential to consider, due to the unorthodox configuration of the aircraft with aspects such as booms and non-retractable undercarriage. For VTOL, all components were considered as such as no components were thought to be streamlined when positioned perpendicular to the free stream. Using a table of C_{D0} contributions normalised by their frontal area (referred to as $C_{D\pi}$) [1], it was possible to model different components as simpler geometric shapes. During VTOL, the wing, fuselage, horizontal stabiliser and booms were modelled as elliptical cylinders [3]. The resulting C_{D0} contributions during VTOL can be seen in Table 3.

Table 3. Miscellaneous drag sources during VTOL

Component	Modelled as	$C_{D\pi}$ [3]	C_{D0} Contribution
Wing	Cylinder ($t \approx 200\%$ chord)	0.91	0.458
Fuselage	Cylinder	1.2	0.251
Tailplane	Cylinder ($t \approx 200\%$ chord)	0.91	0.221
Propeller	Strut	1.17	3.72×10^{-3}
Booms	Cylinder	1.2	0.044

These were summed together with the value for the rotors from Equation 7 (which gave a contribution of 0.5 to C_{D0}). This is displayed graphically in Figure 2, giving a total $C_{D0} = 1.474$ for VTOL - this translates to a drag of 3.91 N in reality.

Again using Equation 7, the C_{D0} contribution of the pusher propeller during cruise was estimated to be 3.56×10^{-3} . Finally, for cruise, the values obtained were input into Equation 8 along with the interference factor Q_c - these can be seen in Table 4 - giving a result of $C_{D0} = 0.0186$.

$$C_{D0} = \frac{\Sigma (C_{f_c} FF_c Q_c S_{wet_c})}{S_{ref}} + C_{D_{misc}} \quad (8)$$

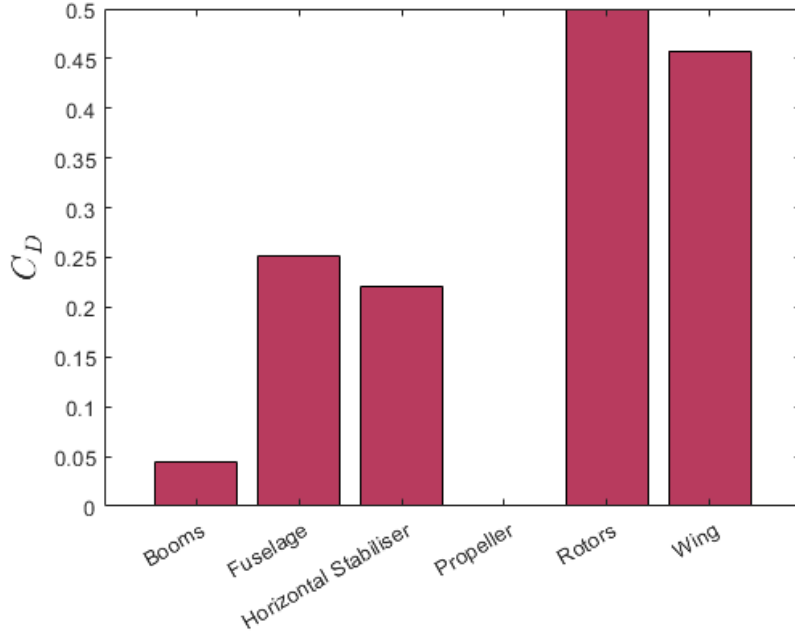


Figure 2. The contributions of all components to VTOL. $C_{D0} = 1.474$, $D = 3.91$ N (Note that the drag contribution of the undercarriage is included in that of the fuselage).

Table 4. Miscellaneous drag sources during VTOL

Component	$C_{fc} \times 10^3$	FF_c	Q_c [1]	C_{D0} Contribution $\times 10^3$
Wing	3.76	1.11	1	8.31
Fuselage	3.22	2.10	1	1.49
Tailplane	4.27	1.10	1.05	2.87
Rotors	3.59	1.01	1.5	0.25
Booms	5.02	1.00	1.5	1.85

3.3 Lift Induced Drag

For the cruise analysis, lift induced drag must also be considered - this is the drag caused by wingtip vortices which induce downwash, deflecting the lift vector. This was omitted for the VTOL analysis as there is no acting lifting surface at that point. The lift induced drag due to the wing and horizontal stabiliser was calculated as in Equation 9 [1].

$$C_{Di} = K [C_{L\alpha} (\alpha + i_w - \alpha_{ow})]^2 + \eta_h \left(\frac{S_h}{S_{ref}} \right) K_h C_{L_h}^2 \quad (9)$$

The tailplane efficiency factor η_h was estimated at 0.9 in Equation 9. Also in this equation, the constants $K = \frac{1}{\pi A R_w e_w}$ and $K = \frac{1}{\pi A R_h e_h}$ had the Oswald efficiencies estimated at $e_w = 0.8$ for the wing and $e_h = 0.95$ for the tailplane [1]. This resulted in a value of $C_{Di} = 6.16 \times 10^{-3}$.

The total breakdown of C_D for cruise is shown in Figure 3, with total $C_D = 0.0248$, which results in a total drag of 105 N during cruise and 47.4 N during loiter. C_D is increased with the payload present by 0.00732. This will be discussed in relation to CFD results in Section 6.2.

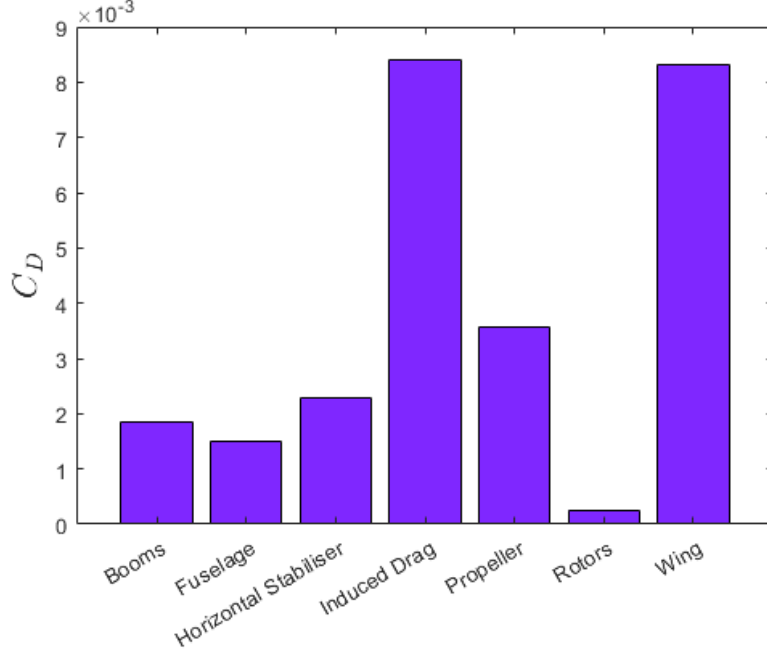


Figure 3. The contributions of all components to cruise/loiter drag. $C_{D0} = 0.0186$, $C_D = 0.0248$, and for cruise $D = 105$ N, while for loiter $D = 47.4$ N.

4 Star-CCM Validation

4.1 Validation against Experimental Data

Prior to using Star-CCM, though it remains an aeronautics industry-standard software, it is essential to ensure that it is valid for our use case. There is no global consensus on how best to validate RANS-based CFD software [5] such as Star-CCM, therefore it was decided to compare the results of an aerofoil with readily available experimental data from wind tunnel tests and 2D Star-CCM simulation. Initially, it was intended to use the same aerofoil as that in the UAV's wing (NACA 6409), however there was a lack of reliable data for this aerofoil. Hence, the NACA 4412 was settled upon, as it has a similar lift curve slope and maximum thickness to the NACA 6409 with reliable experimental data [6].

4.1.1 Mesh Convergence

A 2D polyhedral mesh was set up with a 'prism layer' (a thin, fine, tetrahedral mesh around the edge of the aerofoil) to simulate the likely turbulent boundary layer. A mesh convergence study was run to give an properly converged solution to the simulation, while not using excessive computational power - a finer mesh gives a more accurate result, while a coarser mesh will run faster, so it is necessary to find a compromise between the two by using the largest mesh size that is still converged. This was done using the physical conditions during cruise (namely $V_\infty = 41.1$ m/s, $\rho_{air} = 1.16$ kg/m³ and $p_{atm} = 95460.8$ Pa), along with the K-omega turbulence model. The results of the mesh convergence with regards to the percentage error from minimum mesh base size (an arbitrary overall dimension which aspects of the mesh are built relative in size to) can be seen in Figure 4 and the direct output from the simulations in Figure 5.

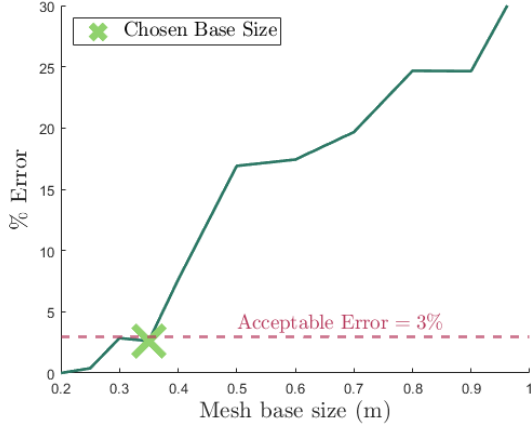


Figure 4. Mesh convergence showing how percentage error to converged solution varies with mesh base size (lower mesh size gives a more accurate result). The chosen base size has less than 3% error.

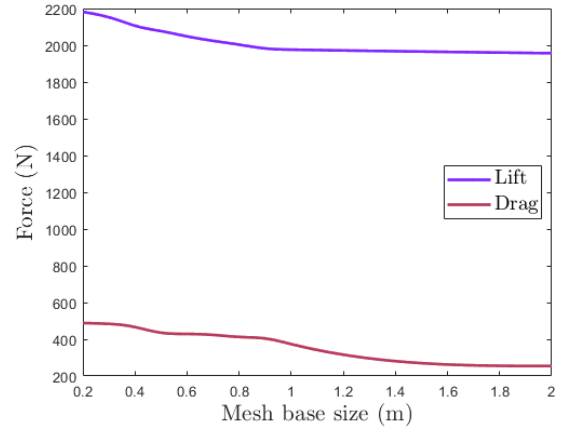


Figure 5. Mesh convergence showing how the aerodynamic forces of lift and drag vary with mesh base size.

As this was merely a validation study, the acceptable error was set at 3% - double what would later be used for the final UAV analysis. This gave a maximum mesh base size of 3.0m to result in a converged solution. This built a mesh with 1281406 cells.

4.1.2 Comparison to Wind Tunnel Data

With a converged mesh, the simulation could be run to find overall lift coefficients and lift-to-drag ratios at varying angles of attack (using the same conditions as the mesh convergence), shown in Figures 6 and 7.

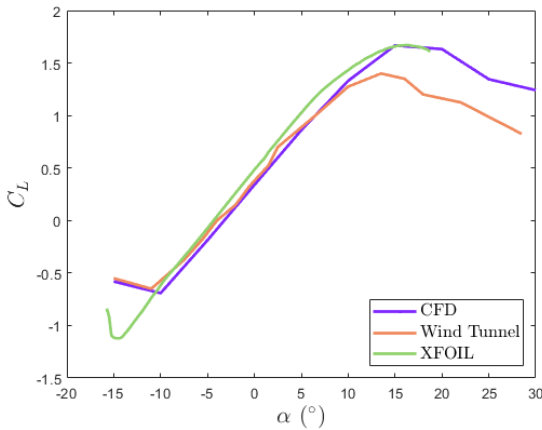


Figure 6. Comparison between lift curve slopes from CFD, wind tunnel experiments [6], and XFOIL.

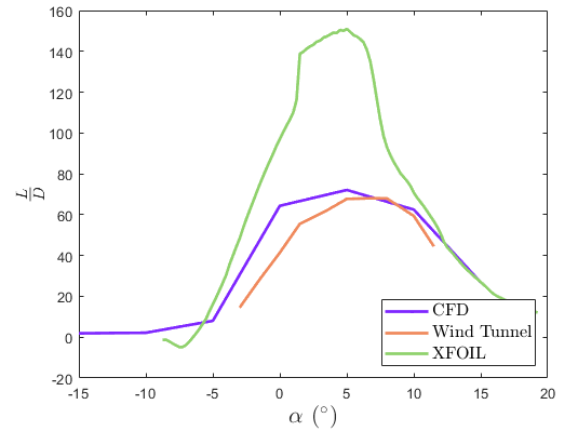


Figure 7. Comparison between lift-to-drag ratios at varying angles of attack from CFD, wind tunnel experiments [6], and XFOIL.

Figure 6 shows an overall good fit between CFD and wind tunnel data. XFOIL was included as a fully inviscid prediction, which can be seen in the stall behaviour at highly negative and highly positive angles of attack where it is a poor match to wind tunnel data. CFD mirrors the experimental results almost perfectly at highly negative and low angles of attack, but deviates slightly prior to and post stall. Figure 7 exposes XFOIL as a highly inaccurate model when drag is included, as the increase in pressure drag due to flow separation at stall is not modelled. CFD closely matches the wind tunnel's

data, which again suggests that the model is valid. This means that it is acceptable for the model to be used in further analyses of our aircraft.

The only issue found with Star-CCM here was in the stall behaviour, which is where separation and the higher proportion of turbulence come into play, suggesting that a better turbulence model is required. Research into this is detailed in the following section.

4.2 Turbulence Model Selection

Turbulent flow is characteristically irregular and impossible to perfectly predict - hence a model must be used to find a solution to the flow. It was decided to use a RANS solver for turbulence, since the alternatives available (direct numerical solutions and scale-resolving simulations) are very computationally expensive and unsuitable for our reasonably high cruise Reynolds number of 1.7×10^6 [7].

While narrowing the selection of turbulence models available, the Reynolds Averaged Navier Stokes equations still require a method of closure - brief summaries of the available ones in Star-CCM are as follows:

- K-Omega: continuity equations for turbulent kinetic energy and *specific* dissipation rate [8].
- K-Epsilon: finds turbulent eddy viscosity by solving continuity equations for *turbulent* kinetic energy and turbulent dissipation rate [9].
- Spalart-Almaras: finds turbulent eddy viscosity by solving one continuity equation for modified diffusivity. [10]
- Reynolds stress with linear pressure strain: Directly solves Reynolds stress tensor by solving their continuity equations [7].
- Reynolds stress with quadratic pressure strain: Same as previous but more precise [7].

As it is not the purpose of the report to discuss the theory behind these models, the continuity equations solved by the K-omega and Reynolds stress turbulence models are shown in Appendix A. Rather than a qualitative analysis, the NACA4412 aerofoil simulation used for validation was reused for the purpose of comparing turbulence models. The simulation was run over a range of angles of attack for different turbulence models, and compared to wind tunnel data and the XFOIL approximation. The results are shown in Figures 8 and 9, although the latter omits wind tunnel data due to the lack of reliable C_D values publicly available.

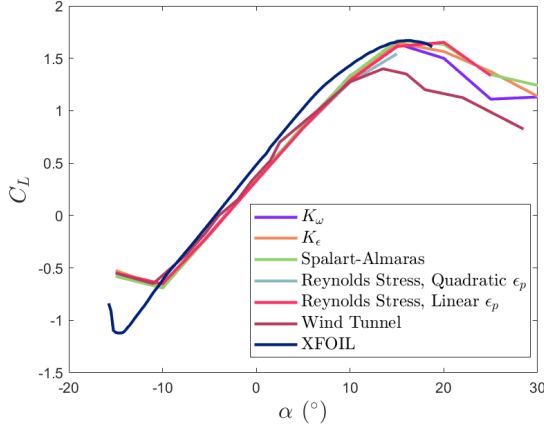


Figure 8. The lift curve slope of the NACA4412 according to the available Star-CCM turbulence models, wind tunnel data, and XFOIL.

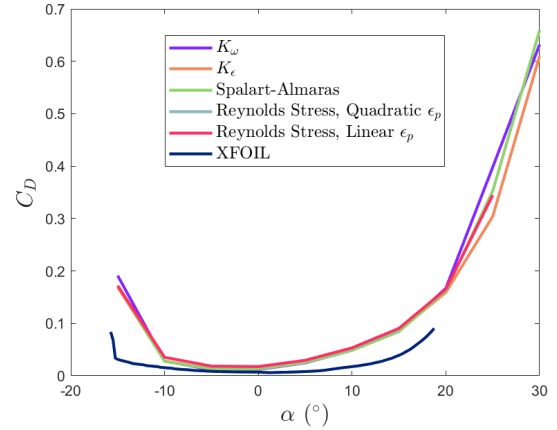


Figure 9. The drag coefficient against angle of attack of the NACA4412 according to the available Star-CCM turbulence models and XFOIL.

With regards to C_D in Figure 9, there is little difference between the different turbulence models - the main outlier is XFOIL, which predictably under-predicts the drag - presumably due to its inviscid nature [11]. Happily, the lift curve in Figure 8 does show differences - XFOIL again fails to capture behaviour at highly negative angles of attack and at stall, however the turbulence models follow the experimental data tightly at negative angles of attack and in the linear lift curve slope region. They deviate at stall, with only the K-omega model providing a reasonable approximation to the wind tunnel results above around 20° . Furthermore, the Reynolds stress models failed to converge at and post stall, indicating that they are unsuitable for this project - a high amount of separation is likely to occur in VTOL, with much of the UAV acting as a bluff body.

The emphasis on separation at VTOL consequently leads to the choice of the K-omega model, as it has the most accurate post-stall behaviour. This decision can be further bolstered by the fact that the main issue with the K-omega model (its high sensitivity to changes in inlet boundary conditions) is moderated within Star-CCM to limit its impact.

5 CFD Modelling of the Whole UAV

It was decided to run a CFD simulation in order to verify analytical calculations against a model that includes the effects of turbulence on a fuselage with unusual airflow interactions between components such as the booms.

5.1 Model Setup

Similarly to the validation setup, a polyhedral mesh was used with a prism layer (as in Figure 10), but this time in three dimensions. A much finer mesh was also needed, as the full aircraft is far more geometrically complex than a simple aerofoil. The UAV geometry used for the mesh was obtained from the Structures team [12] as a CAD model and imported to Star-CCM to be meshed. The rotor and propellers were removed from the CAD model due to the highly computationally expensive analysis of those rotating components [13], and the remaining geometry used for the simulation. The mesh encompassed a $11 \times 4 \times 4$ m block to analyse the flow around the aircraft, which was then reflected across the symmetry axis after the simulation (for faster analysis), shown in Figure 10.

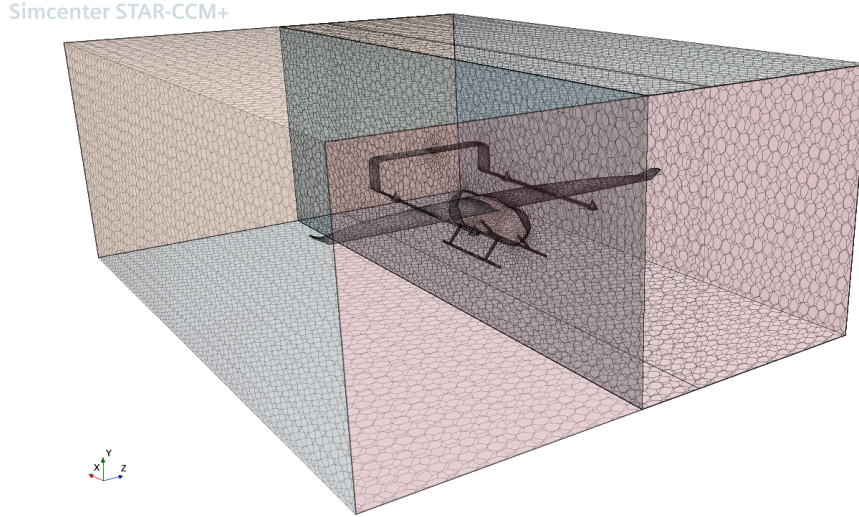


Figure 10. The mesh used for the CFD simulation of the entire vehicle.

5.2 Mesh Convergence

Again, the simulation required use of the largest mesh with a converged solution; this was found using another mesh convergence study in much the same way as Section 4.1.1. The results of this are displayed in Figures 11 and 12, which show the convergence of lift and drag forces respectively against mesh base size.

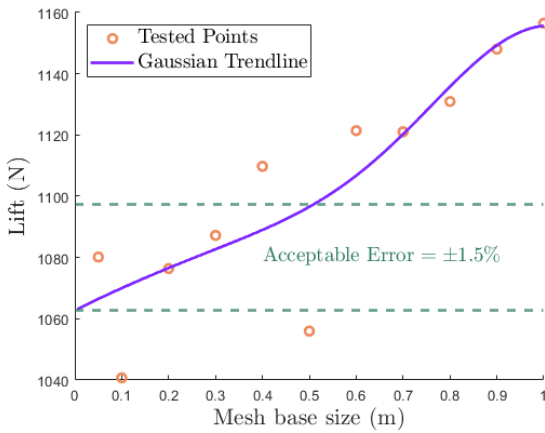


Figure 11. Mesh convergence of lift force in the whole UAV model. The chosen base size was 0.2 m.

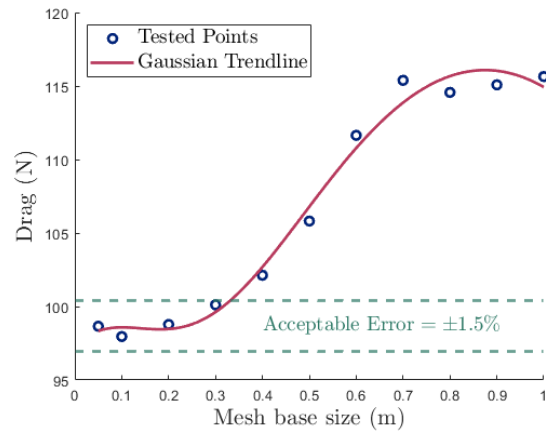


Figure 12. Mesh convergence of drag force in the whole UAV model. The chosen base size was 0.2 m.

In Figures 11 and 12, the expected overall trend of smaller mesh base sizes equating to a lower percentage error to the smallest converged mesh solution - this is illustrated with a Gaussian function trendline. As this simulation is more detailed than the previous 2D validation, a smaller error in convergence was accepted - 1.5% here - that was twice as accurate. This left a base size of 0.3 m, which, for physical reference, is $1/20^{\text{th}}$ of the wing's span.

One piece of information to note from the mesh convergence study is that, while the drag curve seems to fit well with the trend, the lift is susceptible to anomalous results (seen in Figure 11 at 0.5 m and 0.1 m base sizes). This could indicate that the RANS solver did not fully converge at these base sizes, so it was necessary, for the full simulations, to give the solver adequate time to fully converge - not simply level out in a 'false' convergence from which the solver could eventually deviate after

further iterations.

5.3 Results

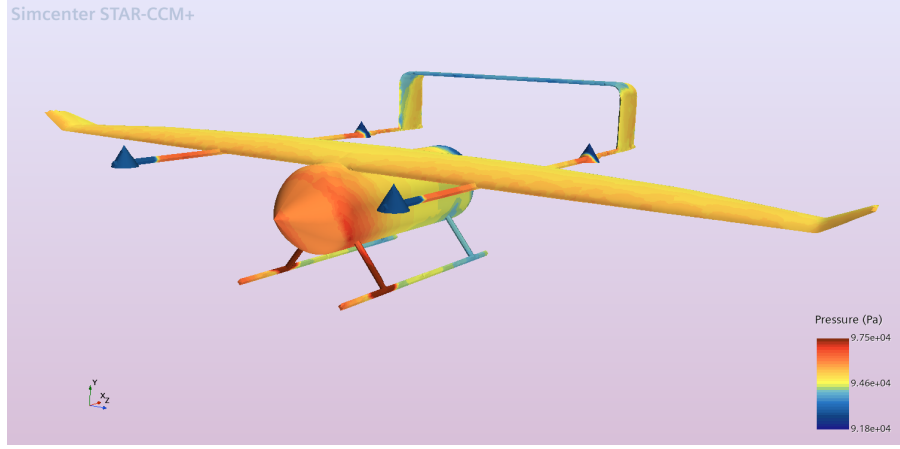


Figure 13. A pressure map of the UAV simulation, included for the reader's interest.

The simulation was run at cruise conditions, with the lift curve slope and drag polar shown in Figures 14 and 15.

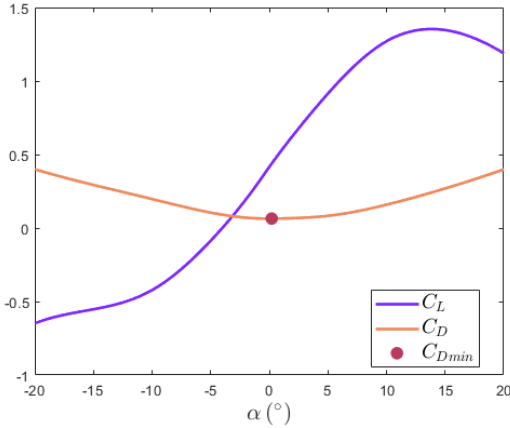


Figure 14. Variation of lift and drag coefficients with angle of attack. C_{Dmin} occurs at $\alpha = 0.20^\circ$.

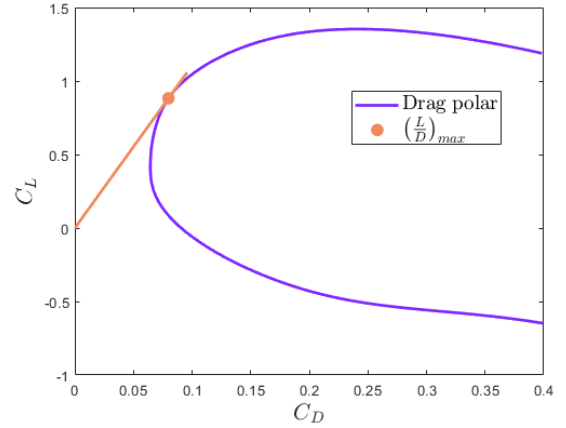


Figure 15. Drag polar of the whole UAV from CFD, with the point of maximum lift-to-drag ratio plotted.

Numerical results for cruise can be seen in Table 5. For the VTOL analysis, the model was rotated 90° and the conditions $V = 1$ m/s and $\rho = 1.225$ kg/m³ used.

6 Discussion

6.1 Comparison of Parasite Drag Approximations

Interestingly, the simple equivalent skin friction method gave a very similar value to the much more complex component build up method - there was a mere 1.61% difference between the two values. This is likely coincidental, as the former method does not take into account the complex (relative to a conventional aeroplane) geometry of our UAV with booms and rotors not being modelled, however it does imply that empirical methods can be as accurate as analytical ones.

Table 5. Key results from the CFD simulation at cruise conditions, compared with previous analytical estimates [13].

Parameter	CFD Value	Analytical Estimate [13]	% Difference
$C_{L\alpha}$	5.72 /rad	5.44 /rad	4.89%
C_{Lmax}	1.35	1.25	7.41%
α_{stall}	13.9°	18.8°	35.5%
α_{minD}	0.202°	0.00°	100%
$(L/D)_{max}$	11.1	15.0	35.1%
$C_{D,cruise}$	0.0644	0.0248	61.5%
$C_{D,VTOL}$	1.56	1.47	5.77%

In general, however, the C_{D0} predicted is quite low (again in comparison to conventional aeroplanes), in spite of aerodynamically inefficient components such as the booms and exposed rotors during cruise. This could reflect the limitations of repurposing methodology intended for more typical aircraft for this use case - though Raymer’s method includes the interference factor Q_c to account for interactions between components [1], the empirical values for this factor may have been underestimated. This could be rectified in later iterations of the aircraft by researching alternative methods of estimating drag for UAVs.

6.2 Comparison of Analytical and Computational Methods

The main shortcoming of analytical methods against CFD is the lack of accurate turbulence modelling, which is instead done using a simple empirical correction. Empirical methods can be accurate, though, since they are derived from historical data [1], though they become less accurate with regards to aircraft more dissimilar to those from which their data is obtained. Since this UAV has an unusual configuration, it may transpire that these methods prove less reliable than a computational RANS solver in the turbulent regime.

The analytical calculations and CFD do, however, agree in case of less turbulent flow - at the angle of attacks with linear lift curve slope (between approximately -7° and 8°). In Table 5, we see an agreement in the values of $C_{L\alpha}$ and C_{Lmax} , which suggests that the lift curve is a good match. The angle of attack at stall, however, has a high 35.5% difference between the analyses. This reinforces the earlier point, as at stall there is higher flow separation than at normal cruise. This is poorly modelled in analytical calculations, while in CFD the K-omega turbulence model is able to more accurately capture the behaviour here.

A design condition that was well met is that of the angle of attack which gives minimum drag (α_{minD}), which showed a small 0.202° deviation from the design constraint to have minimum drag at $\alpha = 0$. This constraint was so that it was possible to loiter with both minimum drag and the payload level, and has been achieved. Any future iterations of the aircraft must ensure that this constraint continues to be met.

Finally, the drag coefficients must be discussed. In VTOL, the C_D of both analysis methods match up well (as demonstrated in Table 5), which suggests that the modelling of major components as stretched cylinders is valid. This accuracy may also be due to the use of experimental data in the modelling of cylinders [3], which will, of course, incorporate the effects of separation better than analytics. The low VTOL speed of 1 m/s may also lend a hand to this, giving an overall far lower Reynolds number than in cruise ($\approx 2 \times 10^5$ vs $\approx 1.7 \times 10^6$) which implies a less turbulent flow [15].

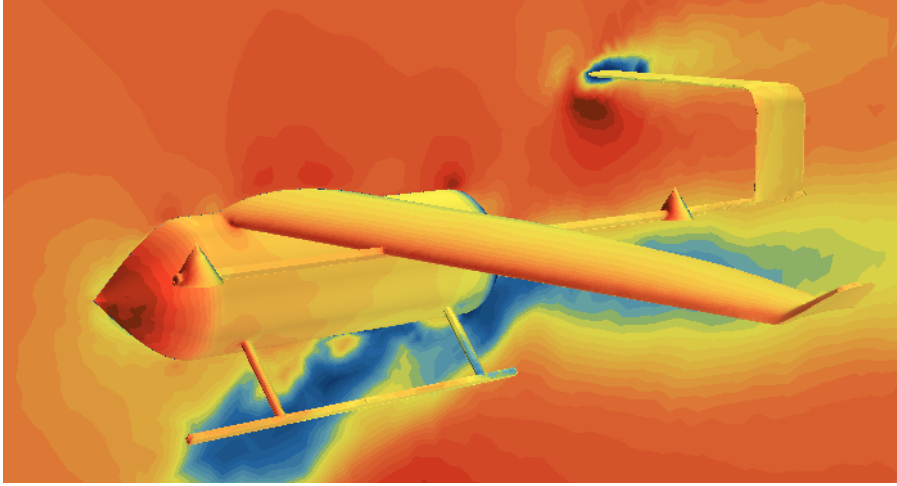


Figure 16. An example of the turbulent regions at highly negative angles of attack.

Cruise drag shows a much higher discrepancy between analytics and CFD with a 61.5% difference in C_{D0} between the two. This brings into question the component build up calculations used [1], which rely heavily on both accurate geometry (in terms of wetted and exposed areas) as well as good choices of empirical values. While, for the former, small errors should not matter excessively, the latter can be difficult to obtain correct values for, since many components of the drone, especially in the early design stages, were not finalised in terms of shape, precise placement, or surface material choice. The under-prediction of C_{D0} does imply that the interference factor, Q_c (as already stated) was improperly considered, with the impact of component interactions underestimated.

This is not to say that CFD is infallible - the validation in Section 4 was done using very simple geometry, so this may not apply to the full UAV. In further iterations of the aircraft (and with fewer time constraints), a more detailed verification of Star-CCM as well as its turbulence models should be completed - this is reinforced by the high levels of turbulence that can be seen on the pressure map in Figure 16.

7 Conclusion

To conclude, it appears overall that analytical calculations are insufficient to analyse the drag on an unusual aircraft configuration, due to significant component interactions that are not modelled by conventional methods. The solution to this issue is likely to use CFD to give a more accurate drag estimate with its ability to solve the entire flow, and simply to use analytics as a baseline to check CFD results against to ensure that they are not completely unrealistic. For future iterations of the aircraft, CFD results should be used to reassess requirements for thrust and other values in light of the higher predicted drag numbers. Furthermore, some prototyping is clearly required so that wind tunnel tests can be conducted and CFD further validated, but this, again, will come along with later iterations of the UAV.

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Appendix

A Additional Details on Turbulence Models

A.1 Continuity/Transport Equations for K-Omega Model

These equations are used by K-omega, K-Epsilon, and Spalart-Allmaras turbulence models.

$$\frac{\partial}{\partial t}(\rho k) + \Delta \cdot (\rho k \bar{\mathbf{v}}) = \Delta \cdot [(\mu + \sigma_k \mu_t) \Delta k] + P_k - \rho \beta^* f_{\beta^*} (\omega k - \omega_0 k_0) + S_k \quad [7] \quad (10)$$

$$\frac{\partial}{\partial t}(\rho \omega) + \Delta \cdot (\rho \omega \bar{\mathbf{v}}) = \Delta \cdot [(\mu + \sigma_\omega \mu_t) \Delta \omega] + P_\omega - \rho \beta f_\beta (\omega^2 - \omega_0^2) + S_\omega \quad [7] \quad (11)$$

where:

$\bar{\mathbf{v}}$ = mean velocity

μ = dynamic viscosity

σ_k, σ_ω = model coefficients

P_k, P_ω = production terms

f_{β^*} = free-shear modification factor

f_β = vortex-stretching modification factor

S_k, S_ω = user-specified source terms

k_0, ω_0 = ambient turbulence values that counteract turbulence delay

A.2 Reynolds Stress Tensor Continuity/Transport Equation

$$\frac{\partial}{\partial t}(\rho \mathbf{R}) + \Delta \cdot (\rho \mathbf{R} \bar{\mathbf{v}}) = \Delta \cdot \mathbf{D} + \mathbf{P} + \mathbf{G} - \frac{2}{3} \mathbf{I} \gamma_M + \underline{\phi} - \rho \underline{\epsilon} + \mathbf{S}_\mathbf{R} \quad [7] \quad (12)$$

where:

ρ = density

$\bar{\mathbf{v}}$ = mean velocity

$\mathbf{D}$ = Reynolds stress diffusion

$\mathbf{P}$ = turbulent production

$\mathbf{G}$ = buoyancy production

$\mathbf{I}$ = identity matrix

γ_M = dilation dissipation

$\underline{\phi}$ = pressure strain tensor

$\underline{\epsilon}$ = turbulent dissipation rate tensor = $\frac{2}{3} \epsilon \mathbf{I}$

$\mathbf{S}_\mathbf{R}$ = user-specified source terms

B Distribution of Work

Topic	People Involved	Person Presenting
Initial Concepts and Concept Selection	All	Parasdeep
Constraint Diagram	Max	Max
Wing Aerofoil Selection	Max	Max
Main Wing Design	Matthew and Max	Matthew
Analytical Drag Estimates	Max	Max
Winglet Design	Naboth	Naboth
Adkins and Leibeck Code	Parasdeep, Naboth, Shahriar	Parasdeep
Propellor Initial Thrust/Power analysis	Parasdeep	Parasdeep
Propeller Design	Parasdeep	Parasdeep
Propeller performance analysis	Parasdeep	Parasdeep
Rotor Initial Thrust/Power analysis	Shahriar	Shahriar
Rotor Design	Shahriar	Shahriar
Rotor Performance Analysis	Shahriar	Shahriar
CFD Setup and Validation for 2D Aerofoil	Matthew, Max	Max
CFD Setup and Validation For Stock Rotor	Matthew, Max	Matthew
Rotor CFD	Shahriar, Matthew	Shahriar
Propeller CFD	Matthew	Matthew
Full Wing CFD	Naboth	Naboth
Full Aircraft CFD	Max	Max
Blanketing Affects of Tailplane	Naboth	Naboth
Tailplane CFD	Naboth	Naboth

Table 6. Separation of Work